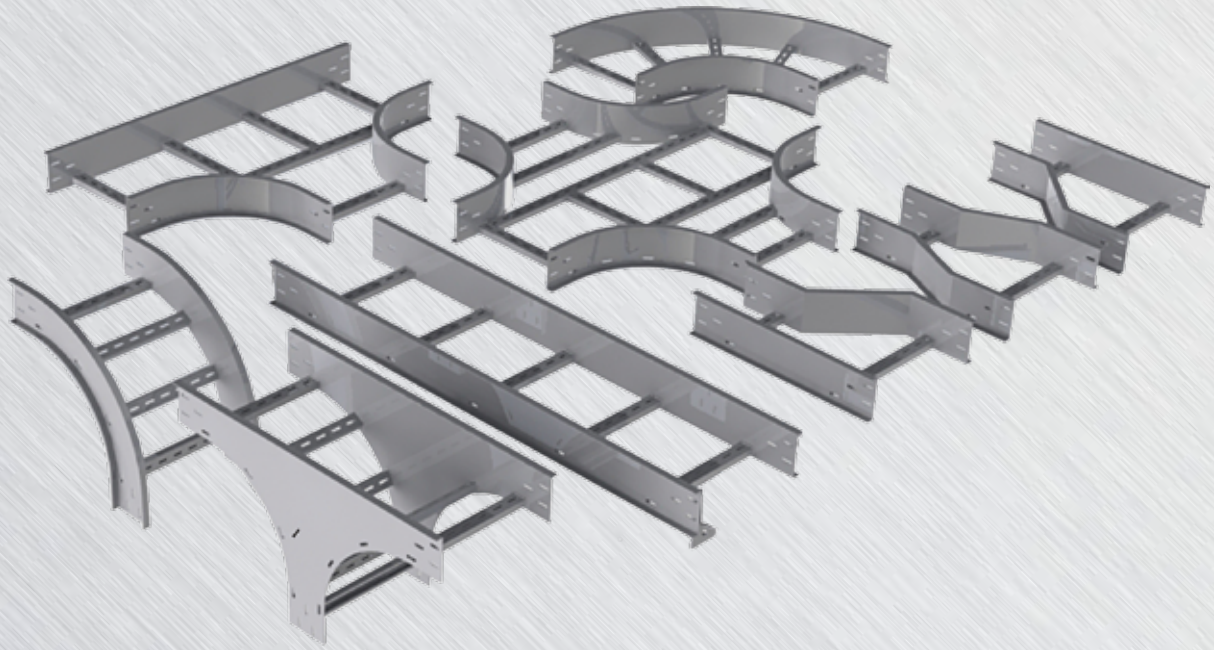
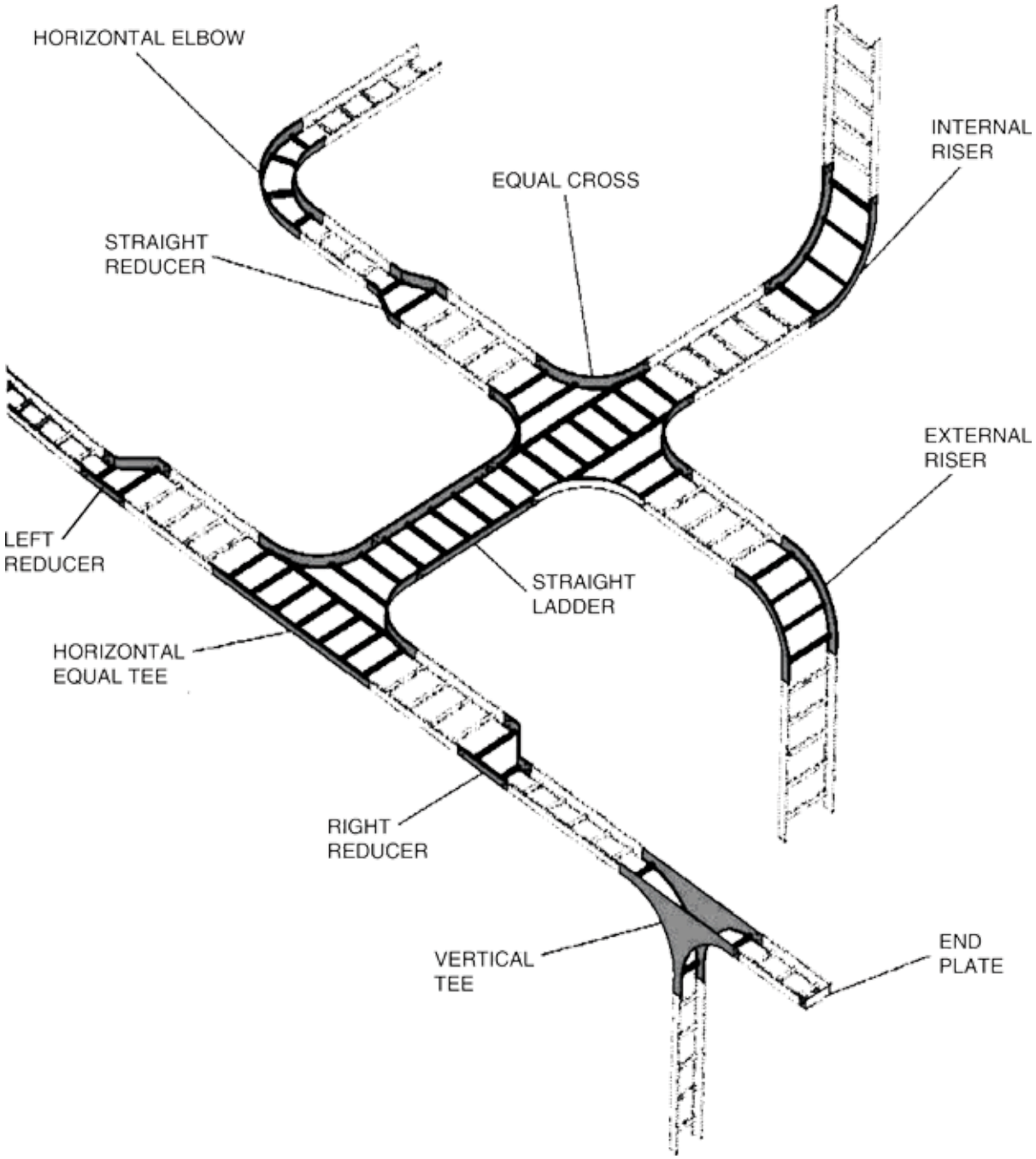


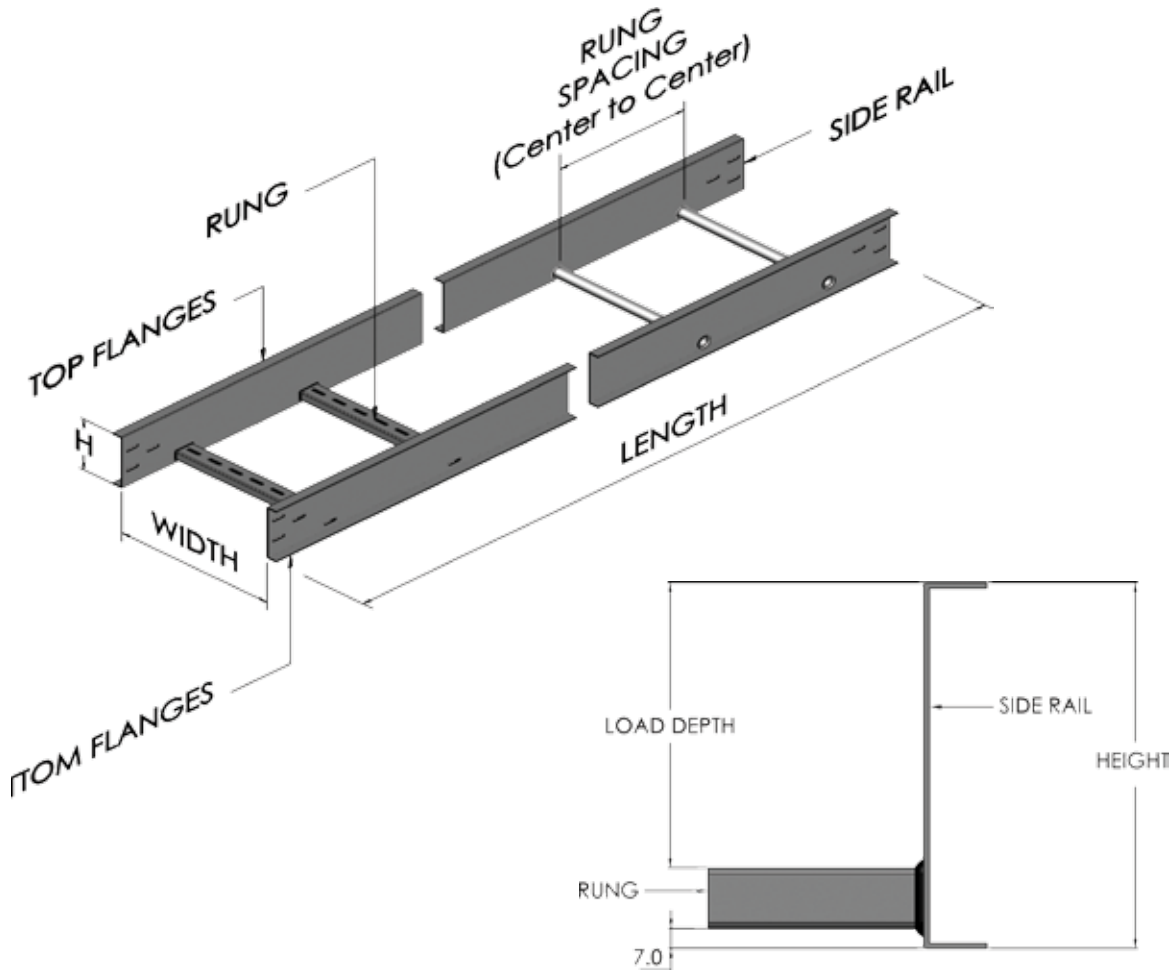
# CABLE LADDER

MODERN METAL INDUSTRIES

# Cable Ladder Typical Layout



# General Technical Information




MMICO has a comprehensive system of cable ladders. MMICO Ladders come in two categories Tubular Swaged Type & C - Channel welded system. MMICO Cable Ladders are comply NEMA VE-1, BS EN ISO 1 461 or ASTM 123 for HDG and for Cold Rolled BS 2994 1976.

Manufactured from a series of welded C-Channel System and Tubular pipe swaged system with standard Height of 110, 135, 160 185mm. (4.25”, 5.25”, 6.25”, 7.25”). Side rail flanges could be inward or outward. Standard length 2.44 to 6.0 M.

## COMPLIANCE STATEMENT

Modern Metal Industries Company never makes any design changes or new products. All products are being manufactured and in compliance with the SAUDI ARAMCO Engineering Standard SAES-P-104 and SAMSS and as well as other related International Standards.

# Technical Specification - Steel Cable Ladder

| Physical Properties, Load and Deflection Data for Steel Ladder | | | | | | | Working (Allowable) Load Capacity-Evenly Distributed Span (ft.) - Simple Beam - Safety Factor : 1.5 | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| NEMA Class Load per Span | | MMICO System Number | Load Depth | Height | Flange | Thickness | 6Ft. (1.8M) | | | 8Ft. (2.4M) | | | 10Ft. (3.1M) | | | 12Ft. (3.7M) | | | 16Ft. (4.9M) | | | 20Ft. (6.1M) | | |
| | | | mm | mm | mm | mm | w | d | k | w | d | k | w | d | k | w | d | k | w | d | k | w | d | k |
| 8A | 8= 8' - 0" | STSS 08A3 | 76 | 108 | 21 | 1.2 | 292 | 0.23 | + | 164 | 0.42 | 0.003 | 103 | 0.63 | 0.006 | 69 | 0.89 | 0.013 | | | | | | |
| | | STSS 08A4 | 102 | 133 | 21 | 1.2 | 383 | 0.18 | + | 210 | 0.31 | 0.001 | 127 | 0.46 | 0.004 | 83 | 0.62 | 0.007 | | | | | | |
| | | STSS 08A5 | 127 | 159 | 21 | 1.2 | 357 | 0.11 | + | 194 | 0.18 | + | 119 | 0.28 | 0.002 | 79 | 0.38 | 0.005 | | | | | | |
| | A=50 lb./ft. | STSS 08A6 | 152 | 184 | 21 | 1.2 | 420 | 0.09 | + | 227 | 0.16 | + | 139 | 0.22 | 0.002 | 92 | 0.30 | 0.003 | | | | | | |
| 8B | (74.4 kg/M) | STSS 08B3 | 76 | 108 | 21 | 1.2 | 292 | 0.23 | + | 164 | 0.42 | 0.003 | 103 | 0.63 | 0.006 | 69 | 0.89 | 0.013 | | | | | | |
| | | STSS 08B4 | 102 | 133 | 21 | 1.2 | 383 | 0.18 | + | 210 | 0.31 | 0.001 | 127 | 0.46 | 0.004 | 83 | 0.62 | 0.007 | | | | | | |
| | B=75 lb./ft. | STSS 08B5 | 127 | 159 | 21 | 1.2 | 357 | 0.11 | + | 194 | 0.18 | + | 119 | 0.28 | 0.002 | 79 | 0.38 | 0.005 | | | | | | |
| | (111.6 kg/M) | STSS 08B6 | 152 | 184 | 21 | 1.2 | 420 | 0.09 | + | 227 | 0.16 | + | 139 | 0.22 | 0.002 | 92 | 0.30 | 0.003 | | | | | | |
| 8C | C=100 lb./ft. | STSS 08C3 | 76 | 108 | 21 | 1.2 | 292 | 0.23 | + | 164 | 0.42 | 0.003 | 103 | 0.63 | 0.006 | 69 | 0.89 | 0.013 | | | | | | |
| | | STSS 08C4 | 102 | 133 | 21 | 1.2 | 383 | 0.18 | + | 210 | 0.31 | 0.001 | 127 | 0.46 | 0.004 | 83 | 0.62 | 0.007 | | | | | | |
| | (148.8 kg/M) | STSS 08C5 | 127 | 159 | 21 | 1.2 | 357 | 0.11 | + | 194 | 0.18 | + | 119 | 0.28 | 0.002 | 79 | 0.38 | 0.005 | | | | | | |
| | | STSS 08C6 | 152 | 184 | 21 | 1.2 | 420 | 0.09 | + | 227 | 0.16 | + | 139 | 0.22 | 0.002 | 92 | 0.30 | 0.003 | | | | | | |
| 12A | 50 lb./ft. | STSS 12A3 | 76 | 108 | 21 | 1.2 | 292 | 0.23 | + | 164 | 0.42 | 0.003 | 103 | 0.63 | 0.006 | 69 | 0.89 | 0.013 | | | | | | |
| | | STSS 12A4 | 102 | 133 | 21 | 1.2 | 383 | 0.18 | + | 210 | 0.31 | 0.001 | 127 | 0.46 | 0.004 | 83 | 0.62 | 0.007 | | | | | | |
| | (74.4 kg/M) | STSS 12A5 | 127 | 159 | 21 | 1.2 | 357 | 0.11 | + | 194 | 0.18 | + | 119 | 0.28 | 0.002 | 79 | 0.38 | 0.005 | | | | | | |
| | 12' span | STSS 12A6 | 152 | 184 | 21 | 1.2 | 420 | 0.09 | + | 227 | 0.16 | + | 139 | 0.22 | 0.002 | 92 | 0.30 | 0.003 | | | | | | |
| 12B | 75 lb./ft. | STSS 12B3 | 76 | 108 | 21 | 1.2 | 360 | 0.26 | + | 198 | 0.46 | 0.002 | 124 | 0.70 | 0.006 | 84 | 0.98 | 0.012 | | | | | | |
| | | STSS 12B4 | 102 | 133 | 21 | 1.2 | 383 | 0.18 | + | 210 | 0.31 | 0.001 | 127 | 0.46 | 0.004 | 83 | 0.62 | 0.007 | | | | | | |
| | (111.6 kg/M) | STSS 12B5 | 127 | 159 | 21 | 1.2 | 357 | 0.11 | + | 194 | 0.18 | + | 119 | 0.28 | 0.002 | 79 | 0.38 | 0.005 | | | | | | |
| | 12' span | STSS 12B6 | 152 | 184 | 21 | 1.2 | 680 | 0.12 | + | 328 | 0.19 | + | 182 | 0.26 | 0.001 | 107 | 0.31 | 0.003 | | | | | | |
| 12C | 100 lb./ft. | STSS 12C3 | 76 | 108 | 21 | 1.5 | 453 | 0.27 | + | 249 | 0.47 | 0.002 | 156 | 0.71 | 0.005 | 106 | 1.00 | 0.009 | | | | | | |
| | | STSS 12C4 | 102 | 133 | 21 | 1.2 | 456 | 0.20 | + | 244 | 0.33 | 0.001 | 152 | 0.51 | 0.003 | 103 | 0.71 | 0.007 | | | | | | |
| | (148.8 kg/M) | STSS 12C5 | 127 | 159 | 21 | 1.5 | 460 | 0.11 | + | 250 | 0.19 | + | 154 | 0.29 | 0.002 | 103 | 0.40 | 0.004 | | | | | | |
| | 12' span | STSS 12C6 | 152 | 184 | 21 | 1.2 | 680 | 0.12 | + | 328 | 0.19 | + | 182 | 0.26 | 0.001 | 107 | 0.31 | 0.003 | | | | | | |
| 16A | 50 lb./ft. | STSS 16A3 | 76 | 108 | 32 | 1.2 | | | | 290 | 0.53 | 0.002 | 182 | 0.81 | 0.004 | 126 | 1.71 | 0.009 | 65 | 1.91 | 0.029 | | | |
| | | STSS 16A4 | 102 | 133 | 32 | 1.2 | | | | 382 | 0.42 | 0.001 | 245 | 0.66 | 0.003 | 170 | 0.94 | 0.006 | 86 | 1.54 | 0.018 | | | |
| | (74.4 kg/M) | STSS 16A5 | 127 | 159 | 32 | 1.2 | | | | 268 | 0.20 | + | 166 | 0.31 | 0.002 | 111 | 0.43 | 0.004 | 58 | 0.71 | 0.012 | | | |
| | 16' span | STSS 16A6 | 152 | 184 | 32 | 1.2 | | | | 322 | 0.17 | + | 199 | 0.26 | 0.001 | 133 | 0.36 | 0.003 | 69 | 0.58 | 0.008 | | | |
| 16B | 75 lb./ft. | STSS 16B3 | 76 | 108 | 32 | 1.5 | | | | 381 | 0.56 | 0.001 | 244 | 0.88 | 0.004 | 169 | 1.27 | 0.008 | 88 | 2.08 | 0.024 | | | |
| | | STSS 16B4 | 102 | 133 | 32 | 1.2 | | | | 382 | 0.42 | 0.001 | 245 | 0.66 | 0.003 | 170 | 0.94 | 0.006 | 86 | 1.54 | 0.018 | | | |
| | (111.6 kg/M) | STSS 16B5 | 127 | 159 | 32 | 1.2 | | | | 426 | 0.31 | + | 259 | 0.46 | 0.002 | 171 | 0.62 | 0.004 | 86 | 0.99 | 0.012 | | | |
| | 16' span | STSS 16B6 | 152 | 184 | 32 | 1.2 | | | | 322 | 0.17 | + | 199 | 0.26 | 0.001 | 133 | 0.36 | 0.003 | 75 | 0.63 | 0.008 | | | |
| 16C | 100 lb./ft. | STSS 16C3 | 76 | 108 | 32 | 1.9 | | | | 487 | 0.58 | 0.001 | 312 | 0.91 | 0.003 | 216 | 1.31 | 0.006 | 113 | 2.16 | 0.019 | | | |
| | | STSS 16C4 | 102 | 133 | 32 | 1.5 | | | | 444 | 0.42 | + | 273 | 0.63 | 0.002 | 186 | 0.89 | 0.005 | 100 | 1.51 | 0.015 | | | |
| | (148.8 kg/M) | STSS 16C5 | 127 | 159 | 32 | 1.5 | | | | 574 | 0.35 | + | 367 | 0.55 | 0.001 | 245 | 0.76 | 0.003 | 123 | 1.22 | 0.010 | | | |
| | 16' span | STSS 16C6 | 152 | 184 | 32 | 1.2 | | | | 504 | 0.25 | + | 306 | 0.37 | 0.001 | 202 | 0.51 | 0.003 | 101 | 0.80 | 0.008 | | | |
| 20A | 50 lb./ft. | STSS 20A3 | 76 | 108 | 32 | 1.5 | | | | | | | | | | 169 | 1.27 | 0.008 | 88 | 2.08 | 0.024 | 52 | 2.98 | 0.057 |
| | | STSS 20A4 | 102 | 133 | 32 | 1.5 | | | | | | | | | | 186 | 0.89 | 0.005 | 100 | 1.51 | 0.015 | 53 | 1.94 | 0.037 |
| | (74.4 kg/M) | STSS 20A5 | 127 | 159 | 32 | 1.5 | | | | | | | | | | 245 | 0.76 | 0.003 | 123 | 1.22 | 0.010 | 66 | 1.59 | 0.024 |
| | 20' span | STSS 20A6 | 152 | 184 | 32 | 1.5 | | | | | | | | | | 178 | 0.38 | 0.002 | 93 | 0.62 | 0.007 | 55 | 0.91 | 0.017 |
| 20B | 75 lb./ft. | STSS 20B3 | 76 | 108 | 38 | 1.9 | | | | | | | | | | 251 | 1.36 | 0.005 | 131 | 2.24 | 0.017 | 77 | 3.23 | 0.042 |
| | | STSS 20B4 | 102 | 133 | 32 | 1.9 | | | | | | | | | | 280 | 1.02 | 0.004 | 145 | 1.67 | 0.012 | 85 | 2.4 | 0.028 |
| | (111.6 kg/M) | STSS 20B5 | 127 | 159 | 32 | 1.5 | | | | | | | | | | 269 | 0.80 | 0.003 | 139 | 1.28 | 0.009 | 81 | 1.82 | 0.022 |
| | 20' span | STSS 20B6 | 152 | 184 | 32 | 1.5 | | | | | | | | | | 291 | 0.59 | 0.002 | 148 | 0.95 | 0.006 | 85 | 1.33 | 0.016 |
| 20C | 100 lb./ft. | STSS 20C3 | 76 | 108 | 38 | 2.5 | | | | | | | | | | 316 | 1.26 | 0.004 | 164 | 2.06 | 0.013 | 100 | 3.09 | 0.013 |
| | | STSS 20C4 | 102 | 133 | 38 | 1.9 | | | | | | | | | | 304 | 1.00 | 0.003 | 157 | 1.00 | 0.010 | 100 | 2.5 | 0.025 |
| | (148.8 kg/M) | STSS 20C5 | 127 | 159 | 32 | 1.9 | | | | | | | | | | 296 | 0.79 | 0.003 | 152 | 0.79 | 0.008 | 105 | 1.91 | 0.018 |
| | 20' span | STSS 20C6 | 152 | 184 | 32 | 1.9 | | | | | | | | | | 379 | 0.66 | 0.002 | 191 | 0.66 | 0.005 | 102 | 1.37 | 0.013 |

to Convert 1.5 safety factor to 2.0 - Multiply w,d,k by - 75

w = Allowable working load (lbs/ft) to convert to kg/M multiply by 148

d = Deflection at midspan (inches) to convert to mm multiply by 25.4.

k = Deflection for each pound of load (inches).

\+ = Deflection is less than 0001/ inch.

# Technical Specification - Aluminum Cable Ladder

| Physical Properties, Load and Deflection Data for Aluminum Ladder | | | | | | | Working (Allowable) Load Capacity-Evenly Distributed Span (ft.) - Simple Beam - Safety Factor : 1.5 | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| NEMA Class Load per Span | | MMICO System Number | Load Depth | Height | Flange | Thickness | 6Ft. (1.8M) | | | 8Ft. (2.4M) | | | 10Ft. (3.1M) | | | 12Ft. (3.7M) | | | 16Ft. (4.9M) | | | 20Ft. (6.1M) | | |
| | | | mm | mm | mm | mm | w | d | k | w | d | k | w | d | k | w | d | k | w | d | k | w | d | k |
| 8A | 8= 8' - 0" A=50 lb./ft. (74.4 kg/M) | STSAL 08A3 | 76 | 108 | 20.6 | 1.5 | 181 | 0.34 | 0.002 | 102 | 0.60 | 0.006 | 65 | 0.94 | 0.014 | 45 | 1.36 | 0.030 | | | | | | |
| | | STSAL 08A4 | 102 | 133 | 20.6 | 1.5 | 324 | 0.33 | 0.001 | 182 | 0.59 | 0.003 | 117 | 0.87 | 0.007 | 77 | 1.26 | 0.016 | | | | | | |
| | | STSAL 08A5 | 127 | 159 | 20.6 | 2.0 | 359 | 0.22 | + | 191 | 0.37 | 0.002 | 115 | 0.55 | 0.005 | 75 | 0.74 | 0.010 | | | | | | |
| | | STSAL 08A6 | 152 | 184 | 20.6 | 1.5 | 416 | 0.18 | + | 221 | 0.31 | 0.001 | 133 | 0.87 | 0.003 | 87 | 0.62 | 0.007 | | | | | | |
| 8B | B=75 lb./ft. (111.6 kg/M) | STSAL 08B3 | 76 | 108 | 20.6 | 1.5 | 181 | 0.34 | 0.002 | 102 | 0.60 | 0.006 | 65 | 0.94 | 0.014 | 45 | 1.36 | 0.030 | | | | | | |
| | | STSAL 08B4 | 102 | 133 | 20.6 | 1.5 | 324 | 0.33 | 0.001 | 182 | 0.59 | 0.003 | 117 | 0.87 | 0.007 | 77 | 1.26 | 0.016 | | | | | | |
| | | STSAL 08B5 | 127 | 159 | 20.6 | 2.0 | 359 | 0.22 | + | 191 | 0.37 | 0.002 | 115 | 0.55 | 0.005 | 75 | 0.74 | 0.010 | | | | | | |
| | | STSAL 08B6 | 152 | 184 | 20.6 | 1.5 | 416 | 0.18 | + | 221 | 0.31 | 0.001 | 133 | 0.87 | 0.003 | 87 | 0.62 | 0.007 | | | | | | |
| 8C | C=100 lb./ft. (148.8 kg/M) | STSAL 08C3 | 76 | 108 | 20.6 | 1.5 | 181 | 0.34 | 0.002 | 102 | 0.60 | 0.006 | 65 | 0.94 | 0.014 | 45 | 1.36 | 0.030 | | | | | | |
| | | STSAL 08C4 | 102 | 133 | 20.6 | 1.5 | 324 | 0.33 | 0.001 | 182 | 0.59 | 0.003 | 117 | 0.87 | 0.007 | 77 | 1.26 | 0.016 | | | | | | |
| | | STSAL 08C5 | 127 | 159 | 20.6 | 2.0 | 359 | 0.22 | + | 191 | 0.37 | 0.002 | 115 | 0.55 | 0.005 | 75 | 0.74 | 0.010 | | | | | | |
| | | STSAL 08C6 | 152 | 184 | 20.6 | 1.5 | 416 | 0.18 | + | 221 | 0.31 | 0.001 | 133 | 0.87 | 0.003 | 87 | 0.62 | 0.007 | | | | | | |
| 12A | 50 lb./ft. (74.4 kg/M) 12' span | STSAL 12A3 | 76 | 108 | 20.6 | 1.5 | 222 | 0.35 | 0.002 | 125 | 0.63 | 0.005 | 80 | 0.99 | 0.012 | 55 | 1.42 | 0.026 | | | | | | |
| | | STSAL 12A4 | 102 | 133 | 20.6 | 1.5 | 324 | 0.33 | 0.001 | 182 | 0.59 | 0.003 | 117 | 0.87 | 0.007 | 77 | 1.26 | 0.016 | | | | | | |
| | | STSAL 12A5 | 127 | 159 | 20.6 | 2.0 | 359 | 0.22 | + | 191 | 0.37 | 0.002 | 115 | 0.55 | 0.005 | 75 | 0.74 | 0.010 | | | | | | |
| | | STSAL 12A6 | 152 | 184 | 20.6 | 1.5 | 416 | 0.18 | + | 221 | 0.31 | 0.001 | 133 | 0.46 | 0.003 | 87 | 0.62 | 0.007 | | | | | | |
| 12B | 75 lb./ft. (111.6 kg/M) 12' span | STSAL 12B3 | 76 | 108 | 20.6 | 2.0 | 354 | 0.45 | 0.001 | 191 | 0.76 | 0.004 | 117 | 1.14 | 0.010 | 78 | 1.57 | 0.020 | | | | | | |
| | | STSAL 12B4 | 102 | 133 | 20.6 | 1.5 | 324 | 0.33 | 0.001 | 182 | 0.59 | 0.003 | 117 | 0.87 | 0.007 | 77 | 1.26 | 0.016 | | | | | | |
| | | STSAL 12B5 | 127 | 159 | 20.6 | 2.0 | 359 | 0.22 | + | 191 | 0.37 | 0.002 | 115 | 0.55 | 0.005 | 75 | 0.74 | 0.010 | | | | | | |
| | | STSAL 12B6 | 152 | 184 | 20.6 | 1.5 | 416 | 0.18 | + | 221 | 0.31 | 0.001 | 133 | 0.46 | 0.003 | 87 | 0.62 | 0.007 | | | | | | |
| 12C | 100 lb./ft. (148.8 kg/M) 12' span | STSAL 12C3 | 76 | 108 | 20.6 | 2.5 | 448 | 0.39 | + | 251 | 0.69 | 0.003 | 161 | 1.08 | 0.007 | 112 | 1.55 | 0.014 | | | | | | |
| | | STSAL 12C4 | 102 | 133 | 20.6 | 2.0 | 394 | 0.31 | + | 222 | 0.55 | 0.002 | 142 | 0.86 | 0.006 | 100 | 1.24 | 0.012 | | | | | | |
| | | STSAL 12C5 | 127 | 159 | 20.6 | 2.0 | 480 | 0.24 | + | 257 | 0.40 | 0.002 | 156 | 0.60 | 0.004 | 102 | 0.81 | 0.008 | | | | | | |
| | | STSAL 12C6 | 152 | 184 | 20.6 | 2.5 | 540 | 0.17 | + | 286 | 0.29 | 0.001 | 172 | 0.43 | 0.003 | 111 | 0.57 | 0.005 | | | | | | |
| 16A | 50 lb./ft. (74.4 kg/M) 16' span | STSAL 16A3 | 76 | 108 | 20.6 | 2.5 | | | | 251 | 0.69 | 0.003 | 161 | 1.08 | 0.007 | 112 | 1.55 | 0.014 | 51 | 2.26 | 0.044 | 25 | 2.75 | 0.110 |
| | | STSAL 16A4 | 102 | 133 | 31.75 | 2.5 | | | | 260 | 0.44 | 0.002 | 166 | 0.69 | 0.004 | 96 | 0.82 | 0.009 | 51 | 1.36 | 0.027 | 28 | 1.83 | 0.065 |
| | | STSAL 16A5 | 127 | 159 | 31.75 | 2.0 | | | | 273 | 0.37 | 0.001 | 155 | 0.51 | 0.003 | 101 | 0.69 | 0.007 | 51 | 1.10 | 0.022 | 24 | 1.28 | 0.053 |
| | | STSAL 16A6 | 152 | 184 | 31.75 | 2.0 | | | | 284 | 0.26 | + | 182 | 0.41 | 0.002 | 110 | 0.52 | 0.005 | 52 | 0.78 | 0.015 | 30 | 1.1 | 0.037 |
| 16B | 75 lb./ft. (111.6 kg/M) 16' span | STSAL 16B3 | 76 | 108 | 31.75 | 2.0 | | | | 342 | 0.72 | 0.002 | 219 | 1.12 | 0.005 | 152 | 1.61 | 0.011 | 82 | 2.74 | 0.033 | 52 | 4.27 | 0.082 |
| | | STSAL 16B4 | 102 | 133 | 31.75 | 3.0 | | | | 349 | 0.51 | 0.001 | 223 | 0.80 | 0.004 | 155 | 1.15 | 0.016 | 79 | 1.85 | 0.023 | 51 | 2.89 | 0.057 |
| | | STSAL 16B5 | 127 | 159 | 31.75 | 1.5 | | | | 380 | 0.49 | 0.001 | 222 | 0.69 | 0.003 | 154 | 1.00 | 0.007 | 78 | 1.61 | 0.021 | 50 | 2.51 | 0.050 |
| | | STSAL 16B6 | 152 | 184 | 38.1 | 1.5 | | | | 377 | 0.32 | + | 241 | 0.50 | 0.002 | 149 | 0.64 | 0.004 | 78 | 1.06 | 0.014 | 50 | 1.65 | 0.033 |
| 16C | 100 lb./ft. (148.8 kg/M) 16' span | STSAL 16C3 | 76 | 108 | 31.75 | 2.5 | | | | 444 | 0.71 | 0.002 | 284 | 0.70 | 0.002 | 188 | 1.53 | 0.008 | 106 | 2.73 | 0.026 | 65 | 4.07 | 0.063 |
| | | STSAL 16C4 | 102 | 133 | 31.75 | 2.0 | | | | 525 | 0.55 | 0.001 | 336 | 0.86 | 0.003 | 233 | 1.24 | 0.005 | 119 | 2.01 | 0.017 | 76 | 3.14 | 0.041 |
| | | STSAL 16C5 | 127 | 159 | 44.45 | 2.0 | | | | 481 | 0.41 | + | 308 | 0.64 | 0.002 | 215 | 0.92 | 0.004 | 121 | 1.63 | 0.013 | 77 | 2.55 | 0.033 |
| | | STSAL 16C6 | 152 | 184 | 38.1 | 1.5 | | | | 542 | 0.44 | + | 318 | 0.62 | 0.002 | 221 | 0.90 | 0.004 | 119 | 1.53 | 0.013 | 76 | 2.39 | 0.032 |
| 20A | 50 lb./ft. (74.4 kg/M) 20' span | STSAL 20A3 | 76 | 108 | 31.75 | 2.0 | | | | | | | | | | 152 | 1.61 | 0.011 | 82 | 2.74 | 0.033 | 52 | 4.27 | 0.082 |
| | | STSAL 20A4 | 102 | 133 | 31.75 | 3.0 | | | | | | | | | | 155 | 1.51 | 0.007 | 79 | 1.85 | 0.023 | 51 | 2.89 | 0.057 |
| | | STSAL 20A5 | 127 | 159 | 31.75 | 1.5 | | | | | | | | | | 154 | 1.00 | 0.006 | 78 | 1.61 | 0.021 | 50 | 2.51 | 0.050 |
| | | STSAL 20A6 | 152 | 184 | 38.1 | 1.5 | | | | | | | | | | 221 | 0.90 | 0.004 | 119 | 1.53 | 0.013 | 76 | 2.39 | 0.032 |
| 20B | 75 lb./ft. (111.6 kg/M) 20' span | STSAL 20B3 | 76 | 108 | 38.1 | 2.0 | | | | | | | | | | 224 | 1.53 | 0.007 | 120 | 2.59 | 0.022 | 77 | 4.05 | 0.053 |
| | | STSAL 20B4 | 102 | 133 | 31.75 | 2.0 | | | | | | | | | | 233 | 1.24 | 0.005 | 119 | 2.01 | 0.017 | 76 | 3.14 | 0.041 |
| | | STSAL 20B5 | 127 | 159 | 44.45 | 2.0 | | | | | | | | | | 215 | 0.92 | 0.004 | 121 | 1.63 | 0.013 | 77 | 2.55 | 0.033 |
| | | STSAL 20B6 | 152 | 184 | 38.1 | 1.5 | | | | | | | | | | 221 | 0.90 | 0.004 | 119 | 1.53 | 0.013 | 76 | 2.39 | 0.032 |
| 20C | 100 lb./ft. (148.8 kg/M) 20' span | STSAL 20C3 | 76 | 108 | 38.1 | 2.5 | | | | | | | | | | 298 | 1.79 | 0.006 | 161 | 3.07 | 0.019 | 103 | 4.79 | 0.047 |
| | | STSAL 20C4 | 102 | 133 | 31.75 | 2.5 | | | | | | | | | | 290 | 1.35 | 0.005 | 156 | 2.30 | 0.015 | 100 | 3.59 | 0.036 |
| | | STSAL 20C5 | 127 | 159 | 44.45 | 2.5 | | | | | | | | | | 316 | 1.23 | 0.004 | 178 | 2.19 | 0.012 | 114 | 3.43 | 0.030 |
| | | STSAL 20C6 | 152 | 184 | 50 | 2.0 | | | | | | | | | | 313 | 0.93 | 0.003 | 176 | 1.66 | 0.099 | 113 | 2.59 | 0.023 |

to Convert 1.5 safety factor to 2.0 - Multiply w,d,k by - 75

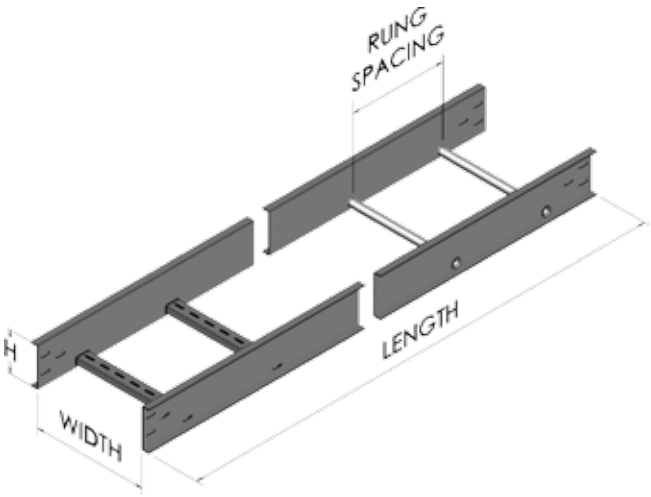
w = Allowable working load (lbs/ft) to convert to kg/M multiply by 148

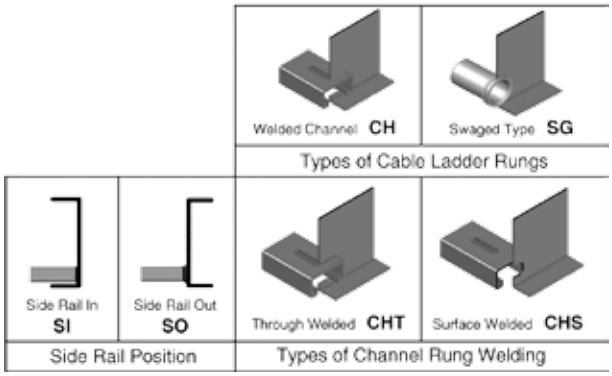
d = Deflection at midspan (inches) to convert to mm multiply by 25.4.

k = Deflection for each pound of load (inches).

\+ = Deflection is less than 0001/ inch.

# Cable Ladder Straight




| Ordering Code | CL | | | | | / | | / | | / | | / | | / | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Cable Ladder | CL | | | | | | | | | | |
| Swaged Type | SG | | | | | | | | | | |
| Welded Channel | CHS | CHT | | | | | | | | | |
| Side Rail In | SI | | | | | | | | | | |
| Side Rail Out | SO | | | | | | | | | | |
| Rung Spacing | mm | 150 | 229 | 250 | 300 | | | | | | |
| | inches | 6” | 9” | 10” | 12” | | | | | | |
| Black Steel | BST | | | | | | | | | | |
| Hot Dip Galvanized after Fabrication | HDG | | | | | | | | | | |
| Hot Dip Galvanized and powder Coating | HDP | | | | | | | | | | |
| Mill Galvanized | MG | | | | | | | | | | |
| Steel Epoxy Coating | EPC | | | | | | | | | | |
| Aluminum | AL | | | | | | | | | | |
| Stainless Steel 304 | S4 | | | | | | | | | | |
| Stainless Steel 316 | S6 | | | | | | | | | | |
| Width (W) | mm | 100 | 150 | 200 | 300 | 450 | 600 | 750 | 900 | 1000 | |
| | inches | 4” | 6” | 8” | 12” | 18” | 24” | 30” | 36” | 40” | |
| Side Rail Height (H) | mm | 110 | 125 | 137 | 150 | 184 | | | | | |
| | inches | 4” | 5” | 5.25” | 6” | 7.25” | | | | | |
| Flange Sizes | mm | 21 | 32 | 38 | 44 | | | | | | |
| | inches | 0.8” | 1.25” | 1.5” | 1.75” | | | | | | |
| Thickness (T) | mm | 1.5 | 2.0 | 2.5 | | | | | | | |
| | inches | 16GA | 14GA | 12GA | | | | | | | |
| Length (L) | mm | 2000 | 2440 | 3000 | | | | | | | |
| | inches | 6.5” | 8” | 10” | | | | | | | |
| NEMA CLASS (N) | 8A | 12A | 16A | 8B | 12B | 16B | 8C | 12C | 16C | | |

**Ordering Code Example :** CLCHSSO300HDG6008/3000/2/32/125/C

**NOTES:**

- Standard pipe size for swaged type cable ladder is 3/4“ for steel & 1“ for aluminum.
- Slotted C-Channel (41x21x2 or 50x25x2) rungs [illegible] for welded type cable lader.
- Special orders can be manufactured upon req[illegible]

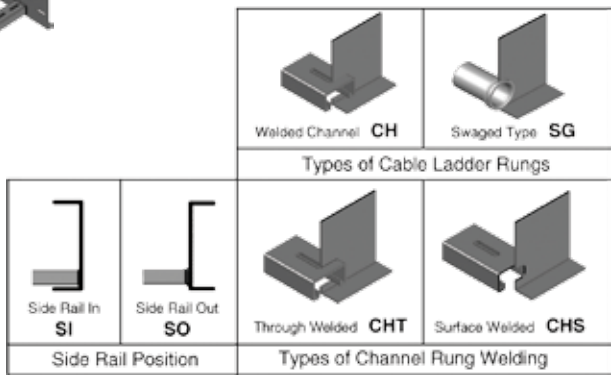
# Cable Ladder Horizontal Elbow



Round Type (R)



Cornered Type (C)




| Ordering Code | CLLH | | | / | | | | | / | | / | | / | | / | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

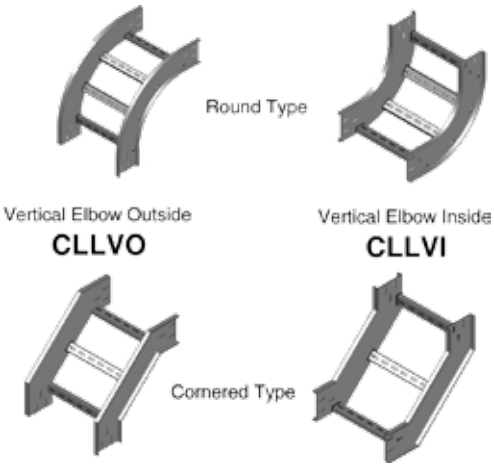
| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Cable Ladder Horizontal Elbow | CLLH | | | | | | | | | |
| Swaged Type | SG | | | | | | | | | |
| Welded Channel | CHS | CHT | | | | | | | | |
| Radius (R) | 300 | 600 | 900 | | | | | | | |
| Angle (A) | 30 | 45 | 60 | 90 | | | | | | |
| Side Rail In | SI | | | | | | | | | |
| Side Rail Out | SO | | | | | | | | | |
| Rung Spacing | mm | 150 | 229 | 250 | 300 | | | | | |
| | inches | 6” | 9” | 10” | 12” | | | | | |
| Black Steel | | | | | BST | | | | | |
| Hot Dip Galvanized after Fabri cation | | | | | HDG | | | | | |
| Hot Dip Galvanized and powder Coating | | | | | HDP | | | | | |
| Mill Galvanized | | | | | MG | | | | | |
| Steel Epoxy Coating | | | | | EPC | | | | | |
| Aluminum | | | | | AL | | | | | |
| Stainless Steel 304 | | | | | S4 | | | | | |
| Stainless Steel 316 | | | | | S6 | | | | | |
| Width (W) | mm | 100 | 150 | 200 | 300 | 450 | 600 | 750 | 900 | 1000 |
| | inches | 4” | 6” | 8” | 12” | 18” | 24” | 30” | 36” | 40” |
| Side Rail Height (H) | mm | 110 | 125 | 137 | 150 | 184 | | | | |
| | inches | 4” | 5” | 5.25” | 6” | 7.25” | | | | |
| Flange Sizes | mm | 21 | 32 | 38 | 44 | | | | | |
| | inches | 0.8” | 1.25” | 1.5” | 1.75” | | | | | |
| Thickness (T) | mm | 1.5 | 2.0 | 2.5 | | | | | | |
| | inches | 16GA | 14GA | 12GA | | | | | | |
| Round Type | R | | | | | | | | | |
| Cornered Type | C | | | | | | | | | |

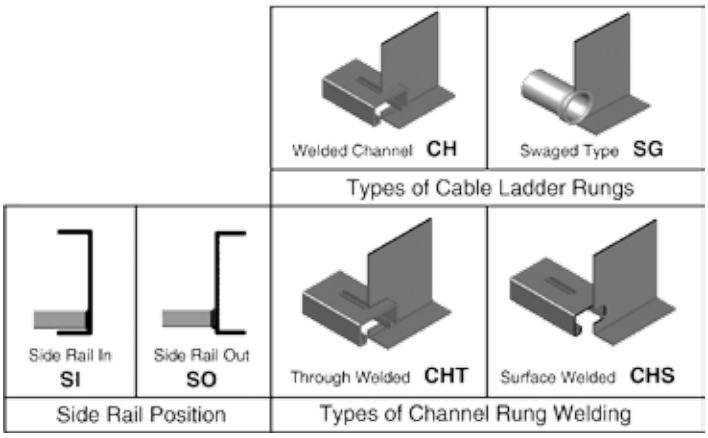
**Ordering Code Example :** CLLHCCHT300/90°SO250HD450/125/32/2/R

**NOTES:**

- Standard pipe size for swaged type cable ladd[illegible] steel & 1“ for aluminum.
- Slotted C-Channel (41x21x2 or 50x25x2) rungs [illegible] for welded type cable lader.
- Special orders can be manufactured upon req[illegible]st.

# Cable Ladder Vertical Elbow




| Ordering Code | CLLV | | | / | | | | | / | | / | | / | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

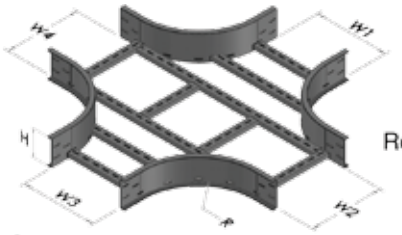
| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Cable Ladder Vertical Elbow | CLLV | | | | | | | | | |
| Swaged Type | SG | | | | | | | | | |
| Welded Channel | CHS | CHT | | | | | | | | |
| Radius (R) | 300 | 600 | 900 | | | | | | | |
| Angle (A) | 30 | 45 | 60 | 90 | | | | | | |
| Side Rail In | SI | | | | | | | | | |
| Side Rail Out | SO | | | | | | | | | |
| Rung Spacing mm | 150 | 229 | 250 | 300 | | | | | | |
| Rung Spacing inches | 6” | 9” | 10” | 12” | | | | | | |
| Black Steel | BST | | | | | | | | | |
| Hot Dip Galvanized after Fabri cation | HDG | | | | | | | | | |
| Hot Dip Galvanized and powder Coating | HDP | | | | | | | | | |
| Mill Galvanized | MG | | | | | | | | | |
| Steel Epoxy Coating | EPC | | | | | | | | | |
| Aluminum | AL | | | | | | | | | |
| Stainless Steel 304 | S4 | | | | | | | | | |
| Stainless Steel 316 | S6 | | | | | | | | | |
| Width (W) mm | 100 | 150 | 200 | 300 | 450 | 600 | 750 | 900 | 1000 | |
| Width (W) inches | 4” | 6” | 8” | 12” | 18” | 24” | 30” | 36” | 40” | |
| Side Rail Height (H) mm | 110 | 125 | 137 | 150 | 184 | | | | | |
| Side Rail Height (H) inches | 4” | 5” | 5.25” | 6” | 7.25” | | | | | |
| Flange Sizes mm | 21 | 32 | 38 | 44 | | | | | | |
| Flange Sizes inches | 0.8” | 1.25” | 1.5” | 1.75” | | | | | | |
| Thickness (T) mm | 1.5 | 2.0 | 2.5 | | | | | | | |
| Thickness (T) inches | 16GA | 14GA | 12GA | | | | | | | |
| Round Type | R | | | | | | | | | |

**Ordering Code Example :** CLLVOCHT300/90SO250HDG450/125/32/R

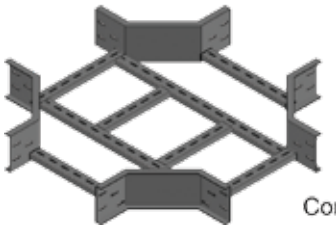
**NOTES:**

- Standard pipe size for swaged type cable ladder is 1¼“ for steel & 1“ for aluminum.
- Slotted C-Channel (41x21x2 or 50x25x2) rungs are used for welded type cable lader.
- Special orders can be manufactured upon request.

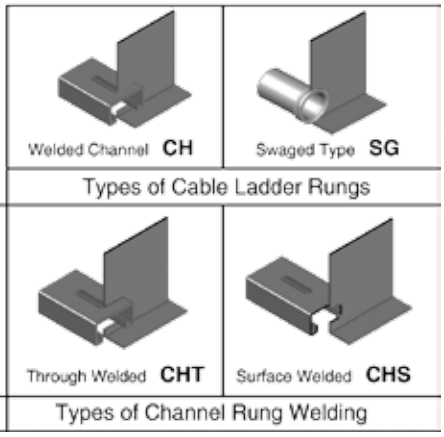
# Cable Ladder Horizontal Cross




Round Type (F



Cornered Type (R)




| Ordering Code | CLX | | | | | / | | / | | / | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|

| | | |
|---|---|---|
| Cable Ladder Horizontal Cross | | CLX |
| Swaged Type | SG | |
| Welded Channel | CHS | CHT |
| Side Rail In | SI | |
| Side Rail Out | SO | |

| | |
|---|---|
| Black Steel | BST |
| Hot Dip Galvanized after Fabri cation | HDG |
| Hot Dip Galvanized and powder Coating | HDP |
| Mill Galvanized | MG |
| Steel Epoxy Coating | EPC |
| Aluminum | AL |
| Stainless Steel 304 | S4 |
| Stainless Steel 316 | S6 |

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Width (W) | mm | 100 | 150 | 200 | 300 | 450 | 600 | 750 | 900 | 1000 |
| | inches | 4” | 6” | 8” | 12” | 18” | 24” | 30” | 36” | 40” |
| Side Rail Height (H) | mm | 110 | 125 | 137 | 150 | 184 | | | | |
| | inches | 4” | 5” | 5.25” | 6” | 7.25” | | | | |
| Flange Sizes | mm | 21 | 32 | 38 | 44 | | | | | |
| | inches | 0.8” | 1.25” | 1.5” | 1.75” | | | | | |
| Thickness (T) | mm | 1.5 | 2.0 | 2.5 | | | | | | |
| | inches | 16GA | 14GA | 12GA | | | | | | |

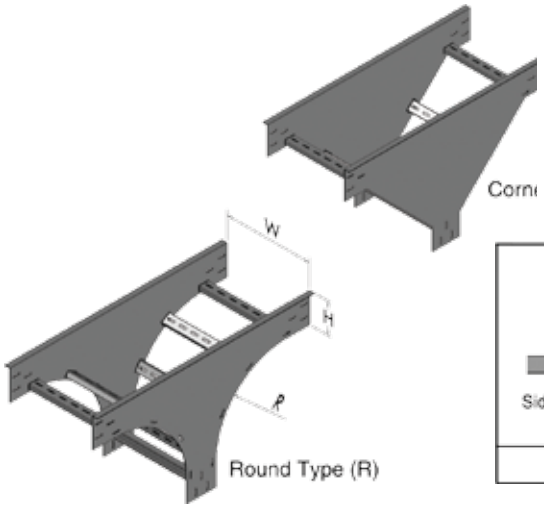
| | |
|---|---|
| Round Type | R |
| Cornered Type | C |

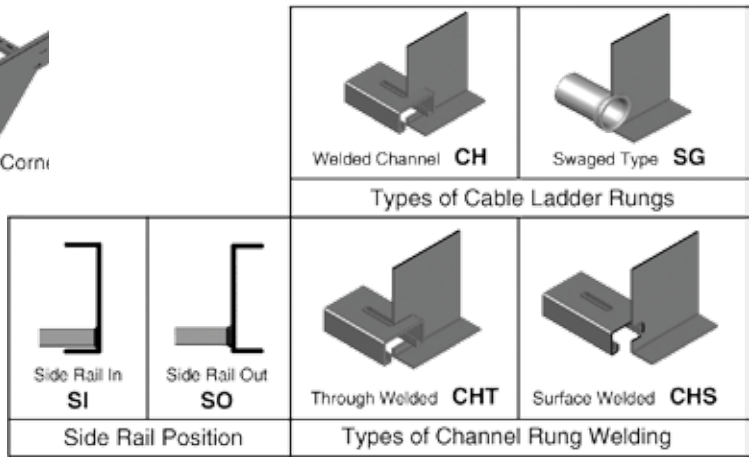
**Ordering Code Example :** CLTHSGSI600HD300/200/300/110/21/2/R

**NOTES:**

- Standard pipe size for swaged type cable ladder is 3/4“ for steel & 1“ for aluminum.
- Slotted C-Channel (41x21x2 or 50x25x2) rungs are used for welded type cable lader.
- Special orders can be manufactured upon request.

# Cable Ladder Vertical Tee




| Ordering Code | CLTV | | | | | / | | / | / | / | |
|---|---|---|---|---|---|---|---|---|---|---|---|

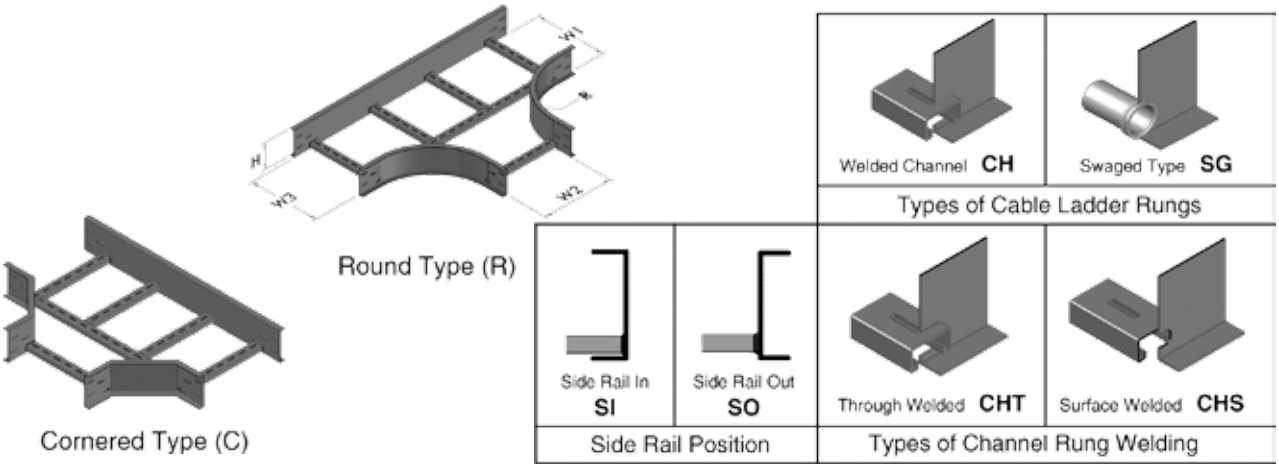
| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Cable Ladder Vertical Tee | | CLTV | | | | | | | | | |
| Swaged Type | SG | | | | | | | | | | |
| Welded Channel | CHS | CHT | | | | | | | | | |
| Side Rail In | SI | | | | | | | | | | |
| Side Rail Out | SO | | | | | | | | | | |
| Black Steel | | BST | | | | | | | | | |
| Hot Dip Galvanized after Fabri cation | | HDG | | | | | | | | | |
| Hot Dip Galvanized and powder Coating | | HDP | | | | | | | | | |
| Mill Galvanized | | MG | | | | | | | | | |
| Steel Epoxy Coating | | EPC | | | | | | | | | |
| Aluminum | | AL | | | | | | | | | |
| Stainless Steel 304 | | S4 | | | | | | | | | |
| Stainless Steel 316 | | S6 | | | | | | | | | |
| Width (W) | mm | 100 | 150 | 200 | 300 | 450 | 600 | 750 | 900 | 1000 | |
| | inches | 4” | 6” | 8” | 12” | 18” | 24” | 30” | 36” | 40” | |
| Side Rail Height (H) | mm | 110 | 125 | 137 | 150 | 184 | | | | | |
| | inches | 4” | 5” | 5.25” | 6” | 7.25” | | | | | |
| Flange Sizes | mm | 21 | 32 | 38 | 44 | | | | | | |
| | inches | 0.8” | 1.25” | 1.5” | 1.75” | | | | | | |
| Thickness (T) | mm | 1.5 | 2.0 | 2.5 | | | | | | | |
| | inches | 16GA | 14GA | 12GA | | | | | | | |
| Round Type | R | | | | | | | | | | |
| Cornered Type | C | | | | | | | | | | |

**Ordering Code Example :** CLTVSGSI600HDP300/200/300/110/21/2/R

**NOTES:**

- Standard pipe size for swaged type cable ladder is 1 1/2“ for steel & 1“ for aluminum.
- Slotted C-Channel (41x21x2 or 50x25x2) rungs are used for welded type cable lader.
- Special orders can be manufactured upon request.

# Cable Ladder Horizontal Tee




| Ordering Code | CLTH | | | | | / | | / | | | / | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|

| | | |
|---|---|---|
| Cable Ladder Vertical Tee | CLTH | |
| Swaged Type | SG | |
| Welded Channel | CHS | CHT |
| Side Rail In | SI | |
| Side Rail Out | SO | |
| Black Steel | BST | |
| Hot Dip Galvanized after Fabri cation | HDG | |
| Hot Dip Galvanized and powder Coating | HDP | |
| Mill Galvanized | MG | |
| Steel Epoxy Coating | EPC | |
| Aluminum | AL | |
| Stainless Steel 304 | S4 | |
| Stainless Steel 316 | S6 | |

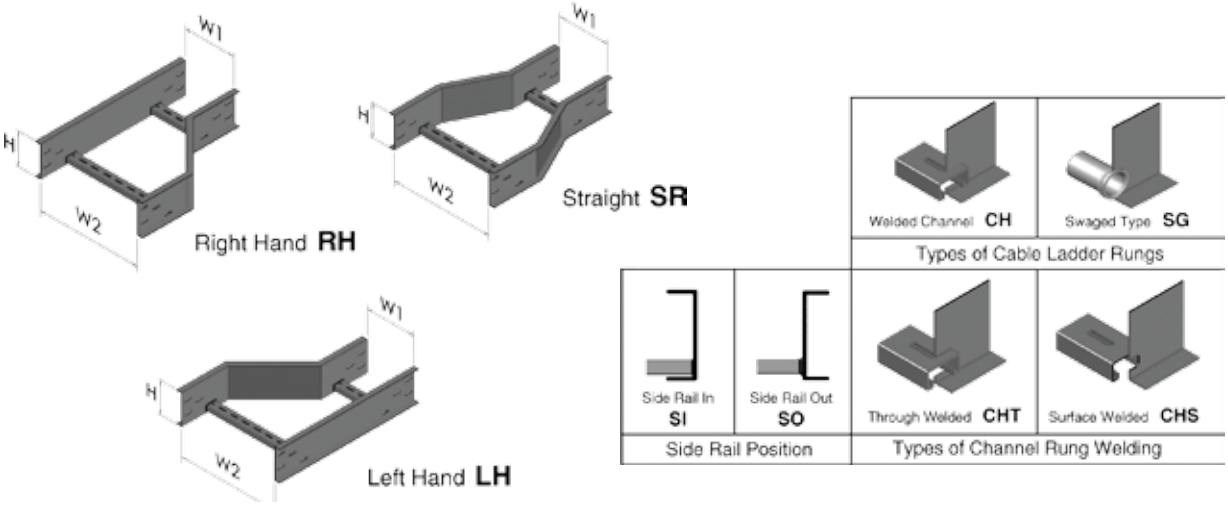
| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Width (W) | mm | 100 | 150 | 200 | 300 | 450 | 600 | 750 | 900 | 1000 |
| | inches | 4” | 6” | 8” | 12” | 18” | 24” | 30” | 36” | 40” |
| Side Rail Height (H) | mm | 110 | 125 | 137 | 150 | 184 | | | | |
| | inches | 4” | 5” | 5.25” | 6” | 7.25” | | | | |
| Flange Sizes | mm | 21 | 32 | 38 | 44 | | | | | |
| | inches | 0.8” | 1.25” | 1.5” | 1.75” | | | | | |
| Thickness (T) | mm | 1.5 | 2.0 | 2.5 | | | | | | |
| | inches | 16GA | 14GA | 12GA | | | | | | |

| | |
|---|---|
| Round Type | R |
| Cornered Type | C |

**Ordering Code Example :** CLXSGSO300HDG450/300/500/200/125/32/2/R

**NOTES:**
- Standard pipe size for swaged type cable ladder is 3/4“ for steel & 1“ for aluminum.
- Slotted C-Channel (41x21x2 or 50x25x2) rungs are used for welded type cable lader.
- Special orders can be manufactured upon request.

# Cable Ladder Reducers




| Ordering Code | CLY | | | | | | / | | / | | / | | / | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| | | |
|---|---|---|
| Cable Ladder Reducer | | CLY |
| Swaged Type | SG | |
| Welded Channel | CHS | CHT |
| Swaged Type | SG | |
| Welded Channel | CHS | |
| Straight | SR | |
| Right Hand | RH | |
| Left Hand | LH | |

| | |
|---|---|
| Black Steel | BST |
| Hot Dip Galvanized after Fabri cation | HDG |
| Hot Dip Galvanized and powder Coating | HDP |
| Mill Galvanized | MG |
| Steel Epoxy Coating | EPC |
| Aluminum | AL |
| Stainless Steel 304 | S4 |
| Stainless Steel 316 | S6 |

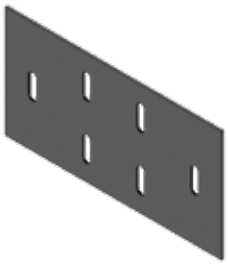
| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Width (W) | mm | 100 | 150 | 200 | 300 | 450 | 600 | 750 | 900 | 1000 |
| | inches | 4” | 6” | 8” | 12” | 18” | 24” | 30” | 36” | 40” |
| Side Rail Height (H) | mm | 110 | 125 | 137 | 150 | 184 | | | | |
| | inches | 4” | 5” | 5.25” | 6” | 7.25” | | | | |
| Flange Sizes | mm | 21 | 32 | 38 | 44 | | | | | |
| | inches | 0.8” | 1.25” | 1.5” | 1.75” | | | | | |
| Thickness (T) | mm | 1.5 | 2.0 | 2.5 | | | | | | |
| | inches | 16GA | 14GA | 12GA | | | | | | |

**Ordering Code Example :** CLYCHTSORHHDG450/200/125/38/2

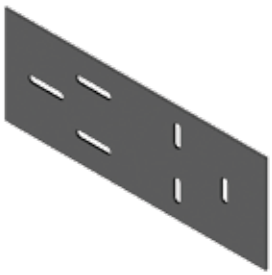
**NOTES:**

- Standard pipe size for swaged type cable ladder is 1 1/2“ for steel & 1“ for aluminum.
- Slotted C-Channel (41x21x2 or 50x25x2) rungs are used for welded type cable lader.
- Special orders can be manufactured upon request.

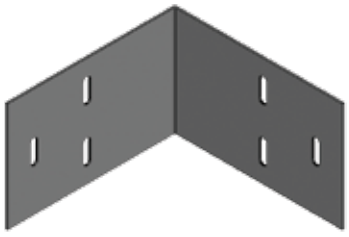
# Cable Ladder Accessories



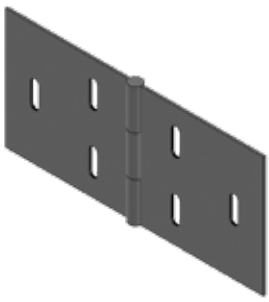
Straight Connector
**(CL01)**



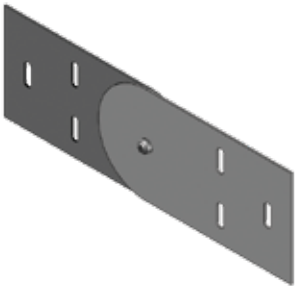
Expansion Connector
**(CL02)**



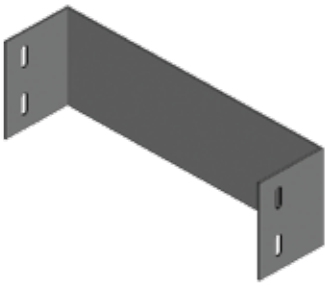
Angle Connector
**(CL03)**



Adjustable Horizontal Joint
**(CL04)**

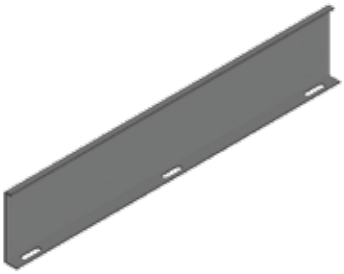


Adjustable Vertical Joint
**(CL05)**

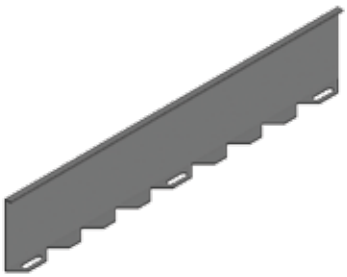


Blind End Plate
**(CL06)**

# Cable Ladder Accessories



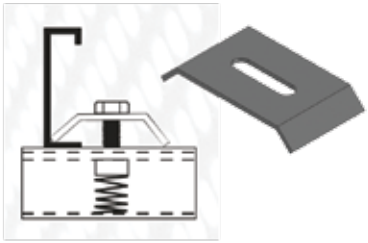
Barrier Strip Straight
**(CL07)**



Barrier Strip Horizontal Fitting
**(CL08)**



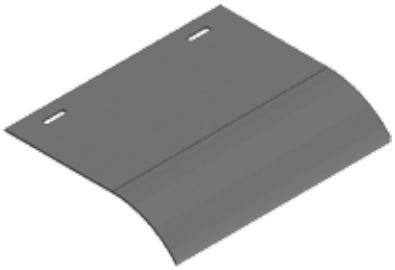
Side Clamp
**(CL09)**



Hold Down Clamp
**(CL10)**



Bonding Jumper
**(CL11)**



Drop Out Plate
**(CL12)**